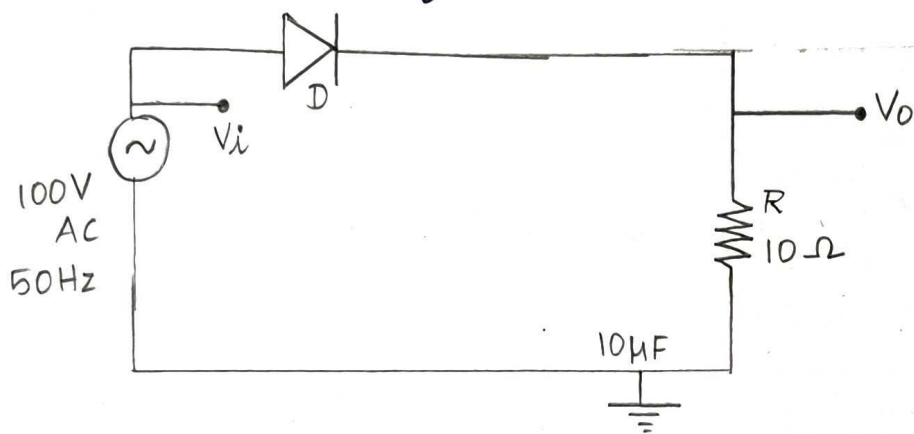
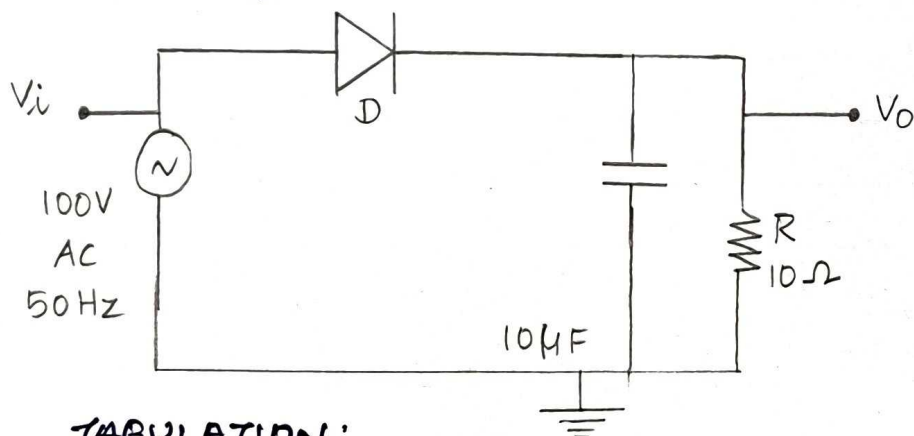


CIRCUIT DIAGRAM HALF RECTIFIER without filter



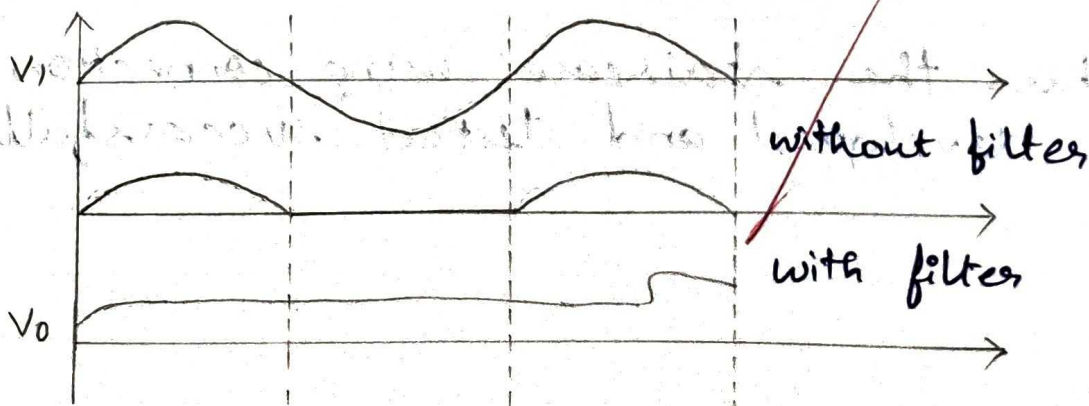
with filter



TABULATION:

Input Voltage		OUTPUT VOLTAGE			
		without filter		with filter	
$V_m(t)$	$t(ms)$	$V_m(v)$	$t(ms)$	$V_m(v)$	$t(ms)$
100V	20ms	100V	100ms	100v, 85v	5ms, 25ms

MODEL GRAPH:



HALF WAVE AND FULL WAVE RECTIFIER

AIM :

To simulate the following circuits in proteus

- i) Half wave rectifier
- ii) Full wave rectifier

APPARATUS REQUIRED :

Laptop with proteus software

THEORY :

A rectifier is a circuit that converts alternating current (AC) into direct current (DC).

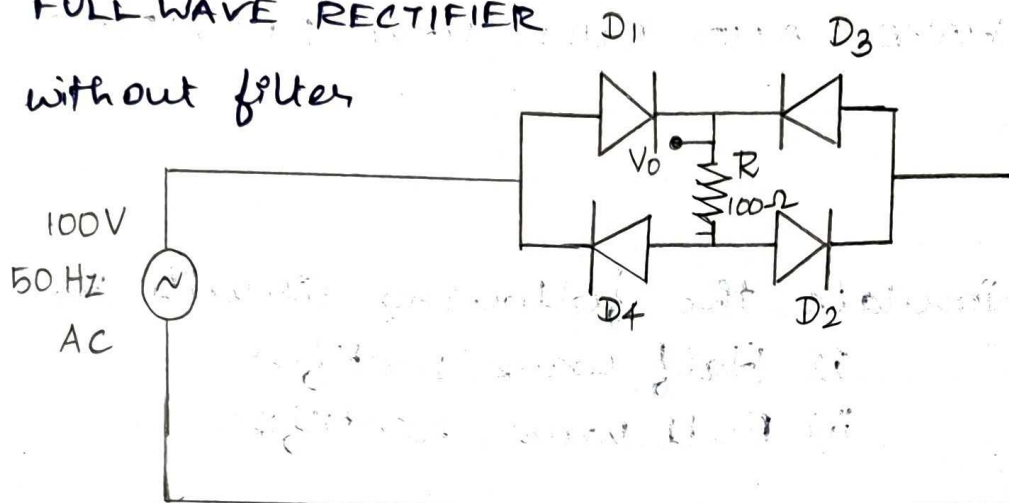
HALF WAVE RECTIFIER :

Halfwave rectifier conducts only during positive half cycles. During negative half cycle the diode conducts and the output voltage is equal to the input voltage. During negative half cycle, the diode does not conduct and the output voltage is equal to zero.

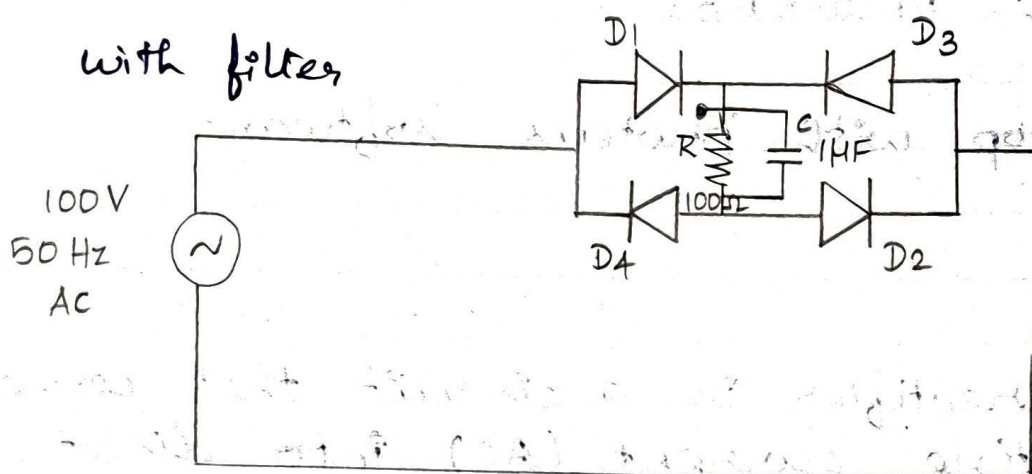
CIRCUIT DIAGRAM:

FULL WAVE RECTIFIER

without filter



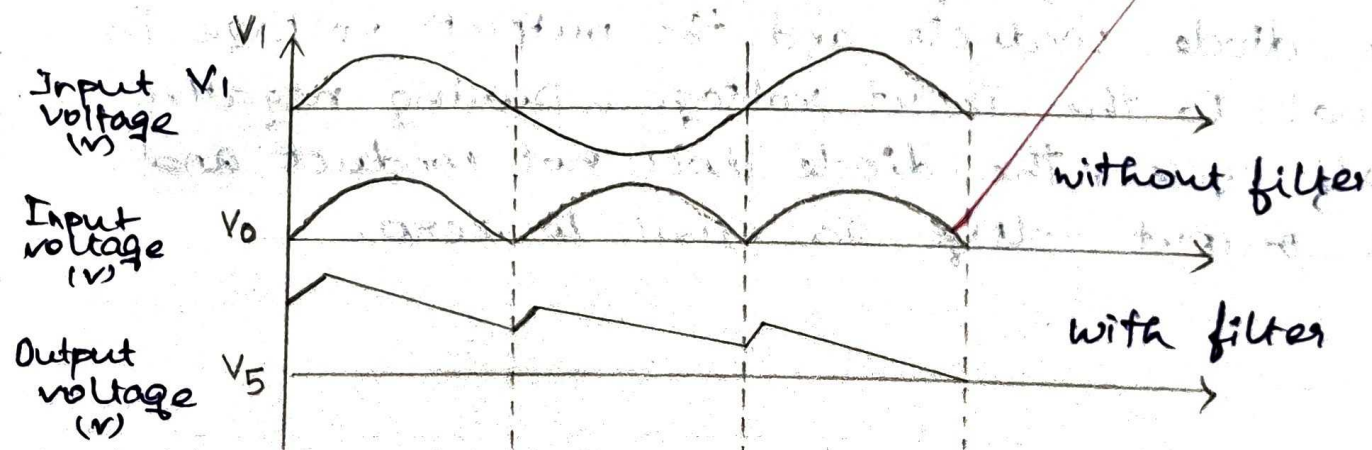
with filter



TABLE

Input voltage		OUTPUT VOLTAGE			
		without filter		with filter	
V_m (V)	t (ms)	V_m (V)	t (ms)	V_m (V)	t (ms)
50 V	20 ms	50 V	5 ms	100 mV, 46 V	5 ms, 14 ms

MODEL GRAPH:



FULL WAVE RECTIFIER:

Full wave rectifier conducts during both positive and negative half cycles. During positive half cycle the output voltage is equal to input voltage, during negative half cycle the output voltage is equal to negative of the input voltage. The output of the rectifier circuit is imp. in pure DC.

In order to get pure DC the output is filtered by a capacitor filter.

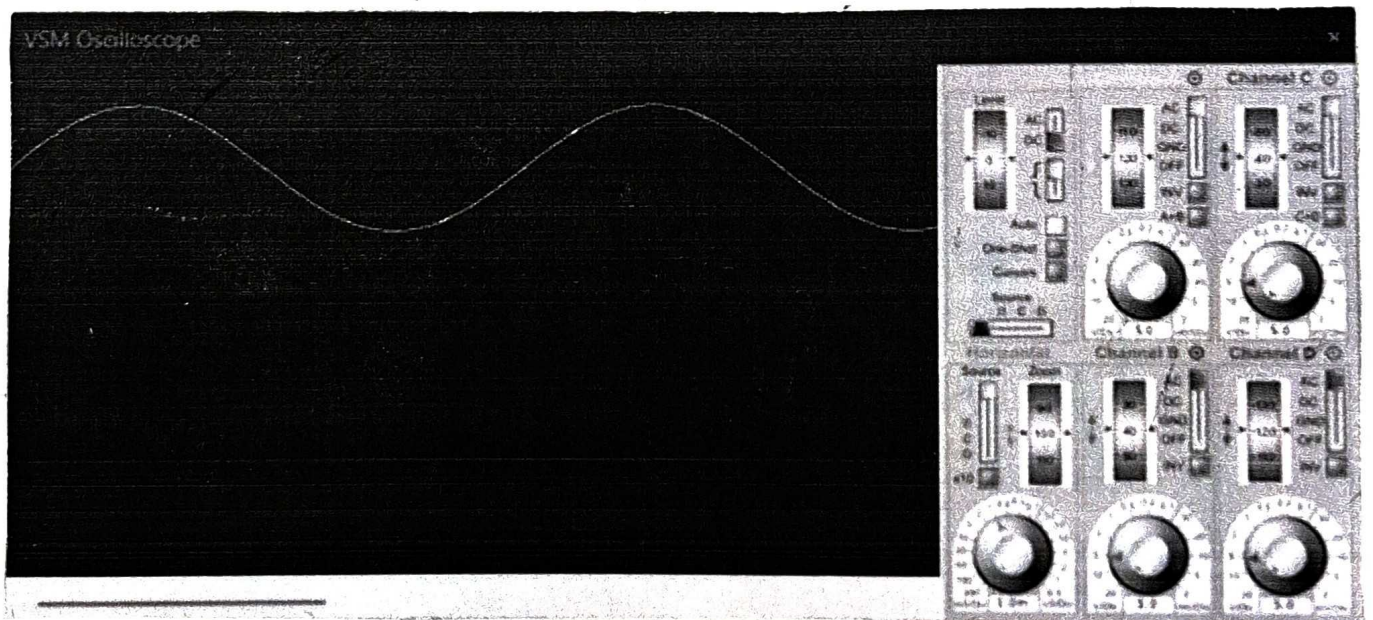
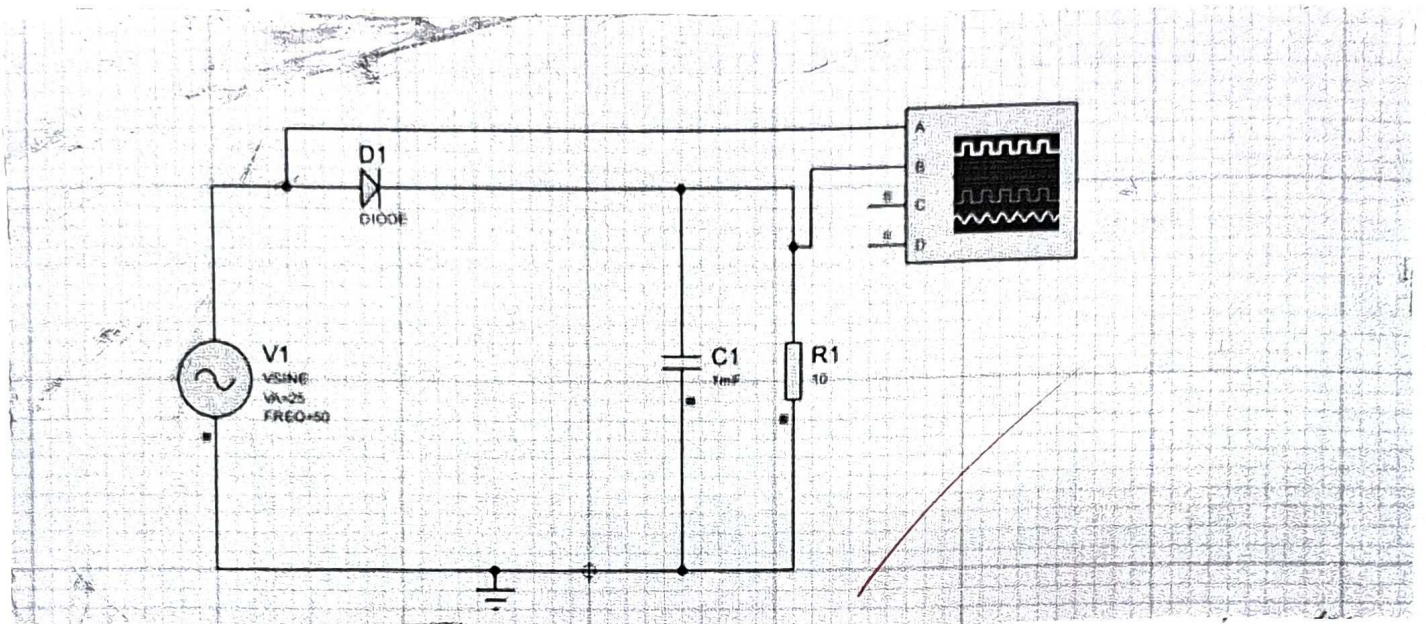
PROCEDURE:

* Drag the required components from the proteus library.

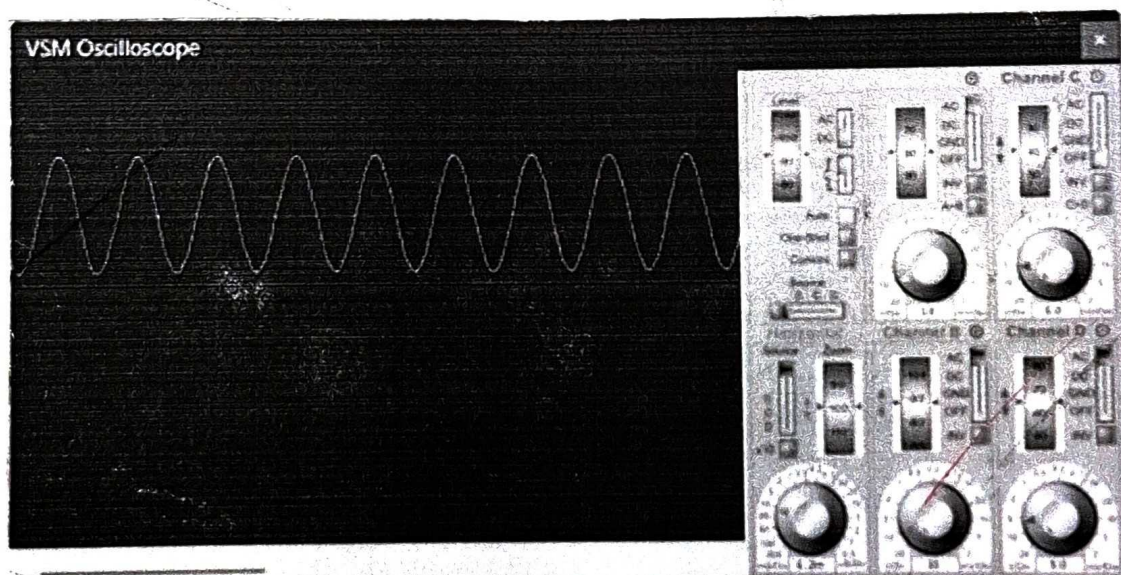
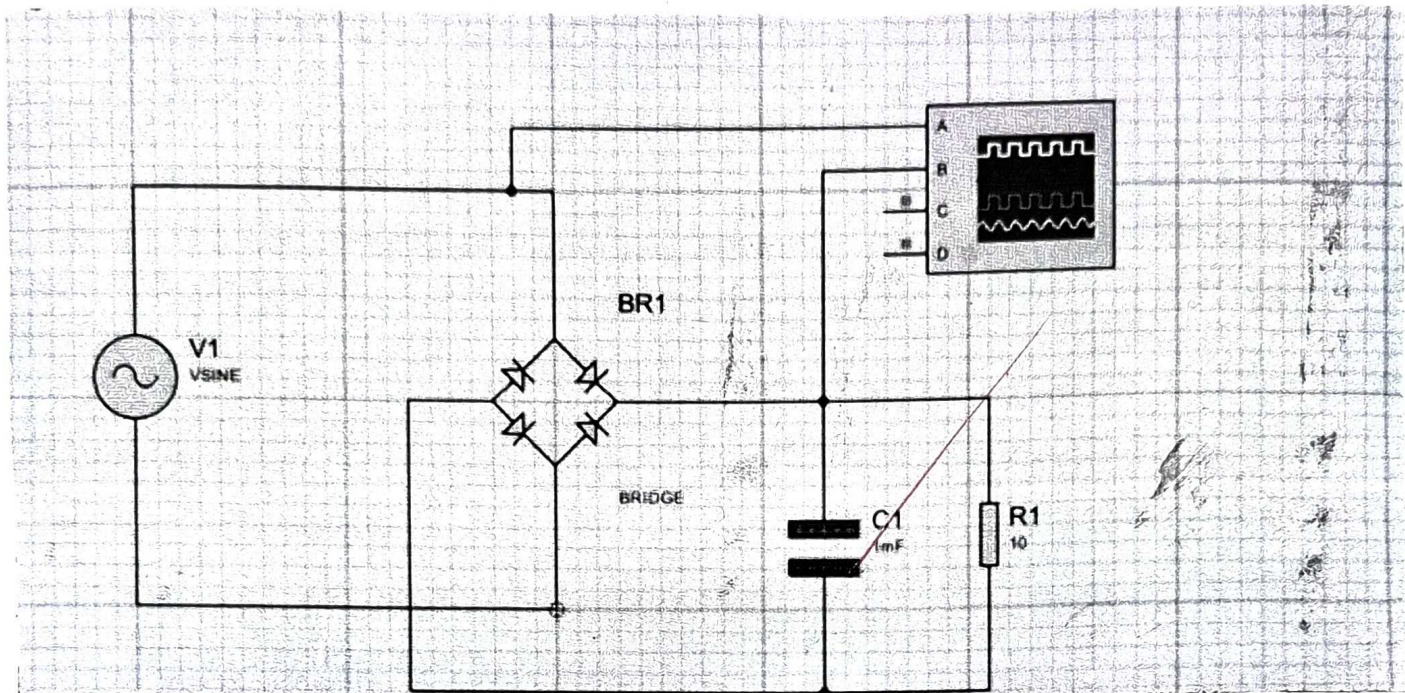
* Connect the components as per the circuit diagram.

* Measure the input and output voltages and note in tabulations.

* Draw the graph for the input and output voltages.

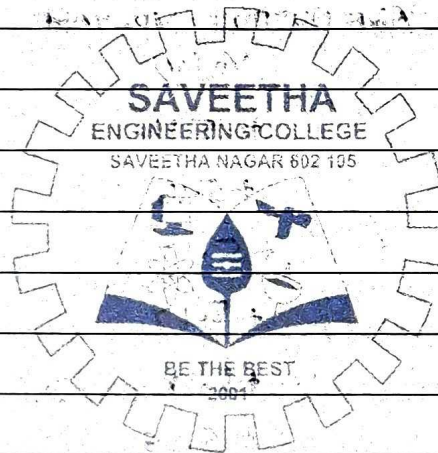


Handwritten notes at the bottom of the page, partially obscured by a red line, include the words "oscilloscope" and "waveform".



99

23/03/20

**RESULT:**

Thus the simulation of the full wave and halfwave rectifier is executed successfully.